Part-A

Birla Institute of Technology & Science, Pilani
II[nd] Semester 2012-13
EEE F244/INSR F244Microelectronic Circuits
Comprehensive Examination (Closed-Book)

MM: 82 **Time: 2 Hrs 15 Min** **Date: 04/5/2013 (FN)**

- Bulk of NMOS connected to ground and bulk of PMOS connected to V_{DD}.
- Specify your assumptions. Justify your answers. Unless specified, assume all MOSFET are biased in saturation region
- Label your sketches properly. All symbols have usual meaning
- ***Unless Given specifically, Take :-*** $V_{DD} = 3.3V$,
  For NMOS device ($\mu_n Cox$ = 140 $\mu A/V^2$, V_T = 0.7 V , λ= 0.1 V^{-1}) and for PMOS device ($\mu_p Cox$ = 40 $\mu A/V^2$, V_T = - 0.8 V , λ= 0.2 V^{-1}).
- All parts of the same question should be done together. Marks will be deducted for wrong units. Your answers must be justified with proper reasons.

Q1(A): For the circuit shown in figure 1, $|V_A|$= 100 V and $V_{CE(Sat)}$ = 0.2 V (for all transistors), I_{REF}= 100 μA , transistors Q3, Q4, Q5 and Q6 has $\beta=\infty$, while transistors Q1 and Q2 have $\beta = 100$, C_L=1pF, R_{sig}=10 KΩ.
(i) Find R_{in} seen at the base of transistor Q1.
(ii) Find the low frequency gain (v_o/v_{sig}).
(iii) Find upper and lower voltage swing at V_o.

(3+3+2=8M)

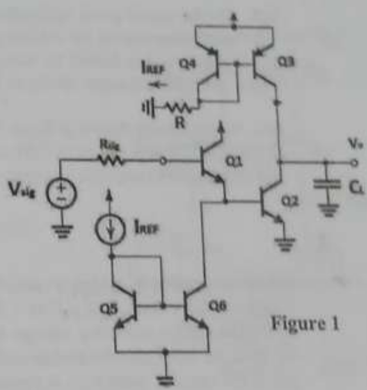

Figure 1

Q1(B): (i) For the circuit shown in figure 2, find I_{o1} and I_{o2} in terms of I_{REF}. Assume all transistors are matched.
(ii) Use this idea to design a circuit that generates currents of 1 mA, 2 mA, and 4 mA using a reference current source of 7 mA assuming $\beta = \infty$. What are the actual values of the currents generated for $\beta = 50$?

(3+3+3=9M)

Q1(C): For the circuit shown in figure 3 assume that for the BJTs β = 100 and $|V_A|$ = 100 V, $|V_{BE}|$ = 0.7 V and MOSFET has W/L=14.3, also I = 100 μA and V_{BIAS} = +1 V. Assume that the output resistance of current source *I* is equal to the output resistance of its connected circuit and current-source *2I* is ideal. For this circuit, calculate:
(i) The voltage at the node between Q_1 and Q_2.
(ii) g_m and r_o for each device.
(iii) The maximum allowable value of v_o.
(iv) The output resistance.
(v) The voltage gain. (v_o/v_i).

(2+2+1+2+4=11M)

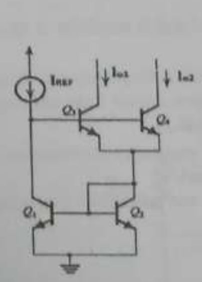

Figure 2

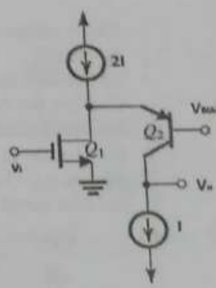

Figure 3

Q2: For the circuit given in figure 4. W_1=W_2=W_3=W_4=W_5 =W_6=**10** μm, L=1 μm. (for all transistors), I_{REF}=50 μA, (Assume:, γ=0)
(i) Calculate equivalent transconductance (G_m), output resistance (R_{out})
(ii) Calculate voltage gain (v_{out}/v_{in}).
(iii) Also calculate upper and lower signal swing.
(iv) If V_{in} and V_b are interchanged than find expression for voltage gain (v_{out}/v_{in}) intuitively.

(4+2+2+3=11M)

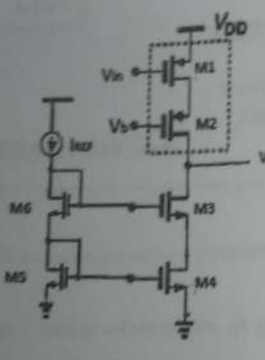

Figure 4

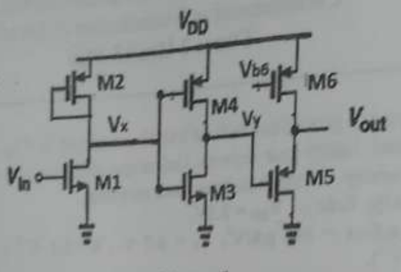

Figure 5

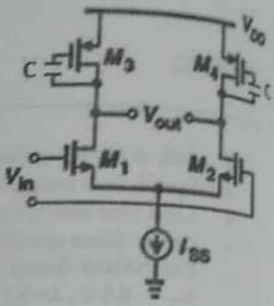

Figure 6

Q3: For the circuit given in figure 5, (Assume:, $\gamma=0$, $V_{DD}=1.8$ V, $|V_{ov,n}|=|V_{ov,p}|$)

(i) Find the expression for voltage gain (v_{out}/v_y), (v_y/v_x), (v_x/v_{in}).

(ii) If this circuit is biased by using a appropriate dc source at V_{in}, then find the value dc bias voltage Vx in order to obtain symmetrical signal swing at Vx. Also calculate upper and lower signal swing at Vx.

(6+4=10 M)

Q4: For the circuit shown in figure 6.

(i) Find out the expression of differential voltage gain $|A_d|$ (intuitively) at very low and high frequency.

(ii)Find out the expression of common mode gain $|A_{CM}|$ (intuitively) at very low and high frequency.

(3+3=6 M)

Q5: It is required to design a cascode amplifier to provide a dc voltage gain of 66 db and utilizing NMOS transistors for which $V_A=10$ V, $W/L=10$, $\mu_nCox = 200\ \mu A/V^2$, $C_{gd}=0.1$ pF, $C_L=1$ pF. Assuming that $R_L=R_{out}$

(i) Determine the overdrive voltage and drain current at which the MOSFETs should be operated. Neglect the body effect.

(ii) Find the unity gain frequency and 3-db frequency.

(iii) If the cascode transistors is removed and R_L remains unchanged. What will the dc gain become ?

(4+4+3=11M)

Q6: A common source amplifier is specified to have $g_m=5$ mA/v, $r_o=40$ kΩ, $C_{gs}=2$ pF, $C_{gd}=0.1$ pF, $C_L=1$ pF, $R_{sig}=20$ kΩ, $R_L=40$ kΩ

(i) Calculate the low frequency small signal voltage gain (A_m) of this amplifier.

(ii) Now use Open circuit time constants to estimate the -3db frequency (f_H) of the amplifier. Hence determine the gain bandwidth product.

(iii) If a 500 Ω resistance is connected in the source lead, find the new values of A_m, f_H and the gain bandwidth product. Assume $g_{mb}=1$mA/V.

(iv) Also sketch and label bode magnitude and phase plot for part (ii) and part (iii) considering only dominant pole.

(3+4+6+3=16M)

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MM: 28

- Bulk of NMO
- Specify your
- Label your sl
- *Unless Given*
- **For NMOS**
  $V_T = -0.8$ V
- All parts of t
  justified with

Q1: For the amplifi
1.4 V and total powe
Given: $(W/L)_{4-7} = 1$0
(i) Calculate the DC
(ii) Find out (W/L) r
(iii) What are minim
(iv) Calculate the vo

Q2: For the amplif
$g_{m1}=g_{m2}=1$mA/V.
(i)Identify the type
(ii)Find feedback f
(iii)Find voltage g
(iv)Find output res

Part-B

Birla Institute of Technology & Science, Pilani
IInd Semester 2012-13
EEE F244/INSR F244 Microelectronic Circuits
Comprehensive Examination (Open book)

MM: 28 **Time: 45 Min** Date: 04/5/2013

- Bulk of NMOS connected to ground and bulk of PMOS connected to V_{DD}.
- Specify your assumptions. Justify your answers. Unless specified, assume all MOSFET are biased in saturation region
- Label your sketches properly. All symbols have usual meaning
- *Unless Given specifically, Take :-* $V_{DD} = 3.3V$,
  For NMOS device ($\mu_n Cox$ = 140 $\mu A/V^2$, V_T = 0.7 V, λ= 0.1 V^{-1}) and for PMOS device ($\mu_p Cox$ = 40 $\mu A/V^2$, V_T = - 0.8 V, λ= 0.2 V^{-1}).
- All parts of the same question should be done together. Marks will be deducted for wrong units. Your answers must be justified with proper reasons.

Q1: For the amplifier circuit shown in figure 1, Overall output voltage swing is 1.4 V and total power dissipation is 5 mW.
Given: $(W/L)_{4-7}$ = 100 and $(W/L)_{M1} = (W/L)_{M2} = (W/L)_{M3}$.
(i) Calculate the DC bias level of the output (V_{out}) and bias voltage V_{b1} and V_{b2}.
(ii) Find out (W/L) ratio of transistor M1.
(iii) What are minimum and maximum allowable input CM level.
(iv) Calculate the voltage gain (V_{out}/V_{in}), common-mode gain and CMRR.

(6+2+4+8=20M)

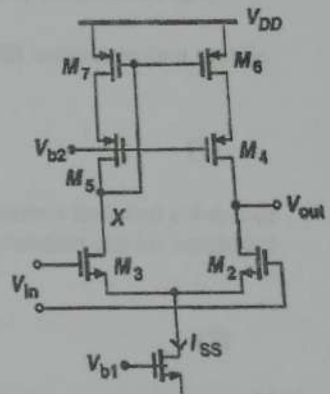

Figure 1

Q2: For the amplifier circuit shown in figure 2, $R_{D1}=R_F=R_{S1}=R_{D2}$=10K, $g_{m1}=g_{m2}$=1mA/V.
(i)Identify the type of feedback topology.
(ii)Find feedback factor β.
(iii)Find voltage gain with feedback ($A_f=V_o/V_i$).
(iv)Find output resistance with feedback (R_{of}).

(1+2+3+2=8M)

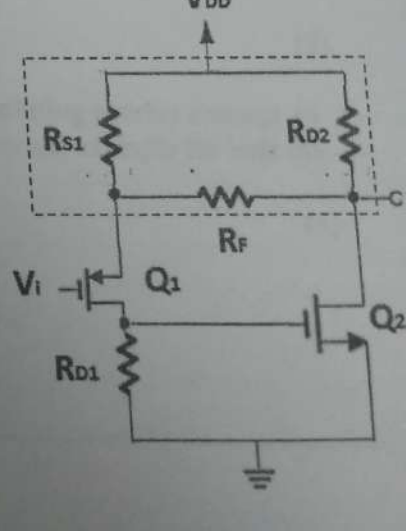

Figure 2

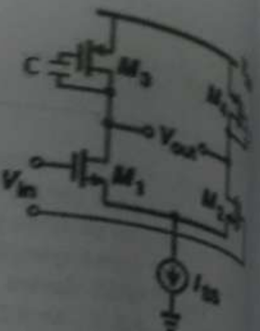

Figure 6

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(6+4=10

w and high frequency.
and high frequency.

(3+3=6

6 db and utilizing NMOS
hat $R_L=R_{out}$
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dc gain become ?

(4+4+3=

pF, C_{gd}=0.1 pF, C_L=1 pF,

e amplifier. Hence determine

$_m$, f_H and the gain bandwidth

nsidering only dominant pole.
(3+4+6+3=

y choosing correct topology and draw

nsistor and calculate the gain.
above, using current mirror load.
e2 and stage3 (A1×A2).(λ_N =0.2 & λ_P

stage to deliver maximum power to

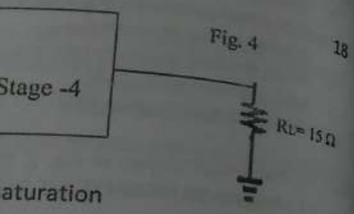

Fig. 4 18

aturation

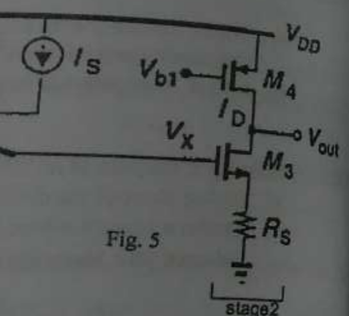

Fig. 5

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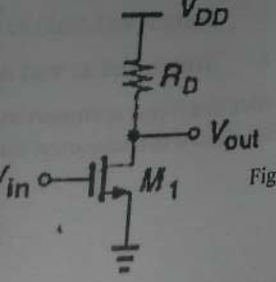


Fig. 6

Birla Institute of Technology and Science, I semester 2012-2013
EEE C424/INSTR C313, Microelectronic Circuits ,
Comprehensive Examination

Total Time: 3 hrs. M.M = 110

Part A (open book)

Max. Marks=25, Time: 40 min.

ID No.

...write to the point answers. No partial marking

1. What is input resistance Rin (looking into the gate) of a diode-connected MOS transistor. Now, what is output resistance Rout of this transistor with source degeneration resistor Rs?

Ans.

2. When a common-drain stage is added between a common-source stage and a load resistor of 50 ohms, what will be the effect on overall voltage gain and why?

Ans.

3. Consider fig.Q_3, Sketch and label Ix versus Vx as Vx varies from 0 to Vdd. Identify important transition points on the sketch.

Ans.

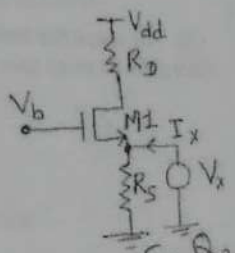


Fig. Q_3

4. Consider a bias voltage generator circuit shown in Fig. Q_4, Determine the value of V_2 and V_{dd} required for this circuit to operate. Given: $\lambda_n = \lambda_p = 0$, $V_{sb} = 0$, I_{bias}=200µA, $(W/L)_1$=5, $(W/L)_2$=3, $(W/L)_3$= 11, $\mu_n C_{ox}$ = 140µA/V^2, V_{tn} = 0.7 V, $\mu_p C_{ox}$ = 40 µA/V^2, V_{tp} = - 0.8 V

Ans.

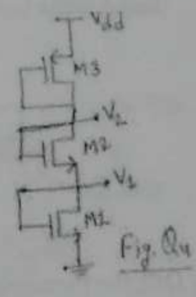


Fig. Q_4

5. Consider the circuit given in fig. Q_5, If the overall capacitance at node V_{out} is dominated by C_L=1pF and if r_o=200kΩ and g_m=600µA/V for all transistors, Determine the value of Rout & hence ω_{-3dB}?

Ans.

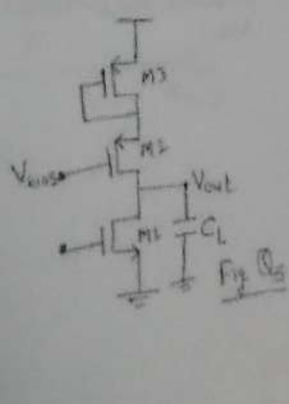


Fig. Q_5

Q6. Consider a circuit shown in Fig. Q_6. Determine the value of R_E and hence find out the maximum value of R_C for the BJT to remain in the active mode. Given: $\beta = 100$, $V_{BE} = 0.7V$, $I_C = 2mA$, $V_{CE(SAT)} = 0.2V$

Ans.

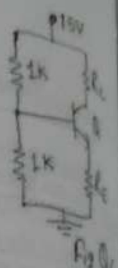


Fig Q_6

Q7.Identify type of feedback for circuit shown in Fig. Q_7.

Ans.

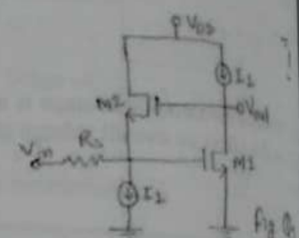


Fig Q_7

Q8. An amplifier having forward gain of 1000 and two poles at 1MHz and 10MHz. calculate the phase margin of a unity gain feedback amplifier.

Q9. Sketch and Label a pMOS cascode amplifier with a complete nMOS cascode current mirror load.Assume I_{ref} of the current mirror is ideal.

Q10. For the cascade amplifier of Question 9, Determine the value of overall output resistance Rout. Given r_o=200kΩ and g_m=600µA/V for all transistors.

Ans.

Instructions-

- State your assumptions clearly
- All parts of the same questio
- with proper reasons.
- Sketches should be properly la
- different transistors
- Take γ=λ=0 in drain current c
- NMOS connected to ground a

Unless mentioned in questioned spe

Unless Given in question , use --

for NMOS device: μ_nCox =

for PMOS device : μ_pCox =

Q1. In multistage amplifier and R_{sig}=20 kΩ. (Given: I_{bias}

a) Calculate the low freq
b) Use the miller Approxi and estimate their val
c) Sketch and label Boo frequencies.
d) Now, using sketch of
e) Redraw the sketches

Q2. For the circuit sho capacitance is CL at ou nearly ideal source.

Using intuitive analys

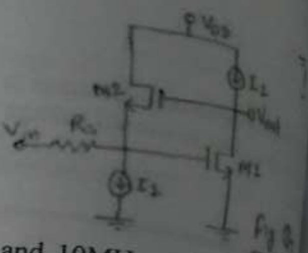
Without Vout1-M7 conne

a) Sketch and label hig model
b) Write the expressio gain (Vout/ Vin)
c) Write the expressi zero frequencies
d) Hence , Write the function (Vout2(s)/

With Vout1-M7 connect

e) Write t

ε and hence find out
Given: $\beta = 100$,



z and 10MHz. calculate the ph

nMOS cascode current mi

output resistance Rout. Gi

Part –B

Max. Marks=85, Time: 2 hr. 20 min.

Instructions-

- State your assumptions clearly. All symbols have their usual meaning. **Question paper has total - 5- questions**
- **All parts of the same question should be done together.**. Marks will be deducted for wrong units. Your answers must be justified with proper reasons.
- Sketches should be properly labeled. Marks would be deducted for calculation mistakes. Use subscripts to distinguish between different transistors
- Take $\gamma=\lambda=0$ in <u>drain current calculations</u> unless necessarily required or specified in the question. Wherever not shown, take bulk of NMOS connected to ground and bulk of PMOS connected to Vdd. Use square law for MOS transistor. Ignore wiring capacitance.

Unless mentioned in questioned specifically, assume all MOS transistors are biased in saturation

Unless Given in question, use -- $V_{DD} = 3.3V$,

for NMOS device: $\mu_nCox = 140\ \mu A/V^2$, $V_T = 0.7$ V, $\lambda = 0.2\ V^{-1}$, Cox= 0.38pF/um^2, Lmin= 1um

for PMOS device: $\mu_pCox = 40\ \mu A/V^2$, $V_T = -0.8$ V, $\lambda = 0.2\ V^{-1}$, C_{OX} =0.38 pF/um^2, Lmin= 1um

Q1. In multistage amplifier shown in fig. Q1, transistor M1 and M3 has C_{gs}=2 pF, C_{gd}=0.1 pF, C_L=1 pF, g_m=5 mA/V and R_{sig}=20 kΩ. (Given: I_{bias}=50 µA, V_{An}= $|V_{Ap}|$= 2 V).

a) Calculate the low frequency small signal voltage gain for the circuit given below in decibels.

b) Use the miller Approximation to calculate number of poles and zeros associated with the circuit intuitively and estimate their values in rad./sec. Include MOS capacitances

c) Sketch and label Bode magnitude and phase plot which should be qualitatively correct using only pole frequencies.

d) Now, using sketch of part(c), approximately calculate Phase Margin (Assume feedback factor=1)

e) Redraw the sketches of part(c), when transmission zero frequencies are included in the plots.

[20]

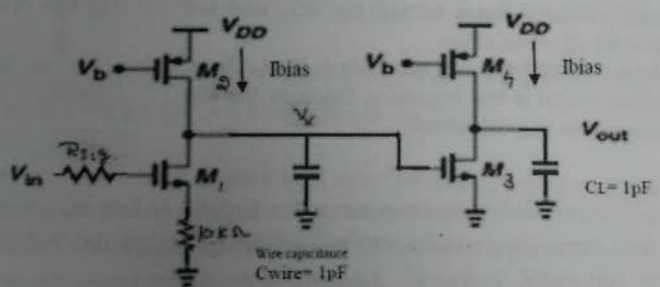


Fig Q1

Q2. For the circuit shown in fig. Q2, load capacitance is CL at output node/ nodes, Iss is nearly ideal source.

Using intuitive analysis--

Without Vout1-M7 connection--- Vout= Vout1-Vout2

a) Sketch and label high frequency small signal model

b) Write the expression of low frequency voltage gain (Vout/ Vin)

c) Write the expression of all pole and transmission zero frequencies

d) Hence , Write the expression for transfer function (Vout2(s)/ Vin(s)]

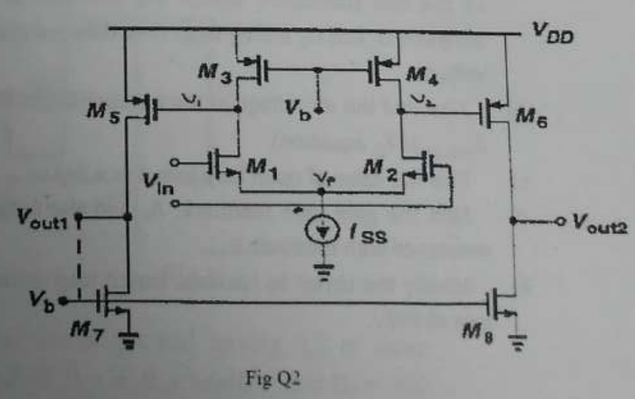


Fig Q2

With Vout1-M7 connection---

e) Write the expression of all pole and transmission zero frequencies

f) Hence , Write the expression for transfer function (Vout2(s)/ Vin(s)]

[13]

Q3. Consider the current mirror load differential amplifier of fig Q3 having trans-conductance mismatches in transistor M1 and M2. Also for differential operation $V_{in1} - V_{in2} = V_{id}$ and for common mode operation $V_{in1} = V_{in2} = V_{icm}$

a) Find the expression for low frequency A_{cm}, A_{dm} and CMRR and then calculate their value using parameters given as -- (r_{o1} and r_{o2} are very high for the input transistors, g_{m1}=1.001 mA/V, g_{m2}=0.999 mA/V, $g_{m3} = g_{m4}$ =1mA/V, $r_{o3} = r_{o4}$= 100 KΩ, r_{o5}= 50KΩ)

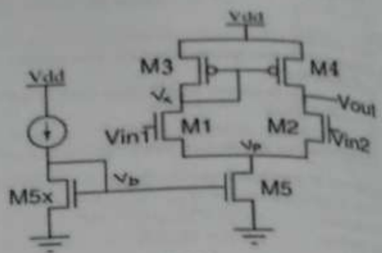


Fig Q3

b) Modify the given circuit to increase CMRR. Justify .
(Note: The bias current is fixed and only one reference current source should be used to bias all the transistors) .

[15]

Q4. For the circuit given in Fig Q4, C1 behaves as short circuit for the signal frequencies of interests. The circuit can accommodate the maximum input voltage level close to V_{DD}.

(Given: The bias current in all transistors are 200 μA. The (W/L)p= 250 and (W/L)n=71.5, assume I_1 and I_2 are nearly ideal current sources). Here assume Vtn= |Vtp |=0.3V

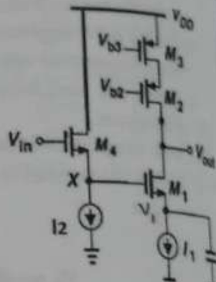


Fig Q4

a) What relationship of gate source voltages of M_1, M_2, M3, and M4 guarantees M_1 in saturation for all values of Vin.
b) Calculate the output signal swing and Maximum allowable dc level at Vx.
c) Sketch and label small signal model at low frequency. Calculate the low frequency small signal voltage gain approximately.

20

Q5. Fig. Q5 shows a series-shunt amplifier with a feedback factor. The amplifier is designed so that v_0 =0 for v_s=0 , with small deviations in v_0 from 0 V dc being minimized by the negative feedback action. The technology utilized has $k_n' = 2\,k_p' = 120 \mu A/V^2$, |Vt|= 0.7V, λ= 0.04 V^{-1} .Assume that the current sources are nearly ideal

a) With the feedback loop opened and the gate terminals of Q_1 and Q_2 grounded, Find the g_m and r_o of each of the five transistors. Ignore the mismatch in I_D between Q_1 and Q_2 arising from their different drain voltages.
b) Also find the dc voltage at the output. (Neglecting λ_{effect} in I_D equation).
c) Find the value of open loop gain A (= v_o/v_s).
d) Find the gain with feedback, A_f, and the output resistance with feedback R_{outf}.
e) Modify the circuit to realize a closed loop voltage gain of 5V/V.

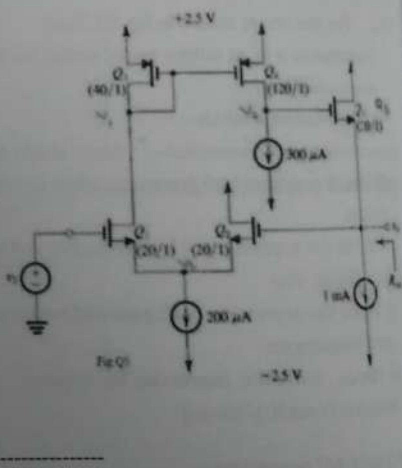


Fig Q5

Birla Institute of Technology and Science, Pilani
EEE F244 / INSTR F244 Microelectronic Circuits
Comprehensive examination II Semester 2014-2015
CLOSED BOOK Max Marks=120

Time: 3 HRS Date: [illegible]

Unless Given specifically
Take — V_{DD} = 3.3V, Lmin=1μm
For NMOS device $\mu_n C_{ox}$ = 140 μA/V², V_t = 0.7 V, λ = 0.1 V⁻¹, C_{ov} = [illegible]
For PMOS device $\mu_p C_{ox}$ = 40 μA/V², V_t = −0.8 V, λ = 0.1 V⁻¹, C_{ov} = [illegible]

NOTE:

If not specified in question——
- ignore γ, λ in drain current equation
- Bulk of nmos connected to ground and bulk of pmos connected to Vdd. Specify your assumptions. Justify your answers.
- Unless specified, assume all MOSFET are biased in saturation region. Label your sketches properly. All symbols have usual meaning. Assume matching of transistors wherever necessary.

Q.1(a) Consider a MOSFET carrying a drain current (I_D) of 1mA with V_{DS} = 0.5 V in active region. Determine the change in I_D if V_{DS} rises to 1V and λ = 0.1 V⁻¹. What is device output impedance 'r_o'? Assuming λ ∝1/L, calculate the change in I_D and output impedance if both Width and Length are doubled.

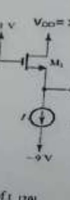


Fig.(1)

Q1(b) For the **Fig.(1)**, V_{GS} = V_{DS} = 3V, use λ = 0.1 V⁻¹
(i) Find the voltage V_2, if the transistor is in saturation at I = I_D = 2mA and V_T = 1V.
(ii) Determine minimum value of drain -source voltage of M1 to remain in saturation region.
(iii) Now a resistor R is inserted between Vdd and drain terminal of M1. For result of part (ii), find the value of R
(iv) using above results, If current source 'I' requires a voltage of 2V across it, determine the value of r that can be inserted between source terminal of M_1 and current source. [20]

Q-2 Consider the circuits given in **Fig(2)**.
a) Redraw the given circuit in **Fig. 2(a)** for very low frequency operation. Find the expression of voltage gain (V_{out}/V_{in}) for the new circuit intuitively.
b) Redraw the circuit in **Fig. 2(a)** for very high frequency operation. Draw small signal model of new circuit and find expression of voltage gain (V_{out}/V_{in}). ignore parasitic capacitances.
c) The circuit shown in **Fig. 2(b)** has a bias current (I_{bias}) of 100μA and W/L values for all the transistors are 10. Calculate values of small signal voltage gain (V_{out}/V_{in}) and output voltage swing. Assume γ=0 and λ≠0

[20]

Fig.2(a)

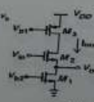


Fig.2(b)

Q-3 For the circuits shown in **Fig.3 (a, b, c, d, e, f g, h, i, j)**, assume all transistors have same g_m and r_o, I_{bias}, W/L and uC_{ox}, V_{ov}. Ignore γ

Answer the following to the point (very brief) (please answer in same sequence)-

a) For **fig. 3a**, write expression for d c voltage at node vout
b) For **fig. 3b**, write the expression for resistance at node vout
c) For **fig. 3c**, write the expression for dc voltage (Vin-Vout)
d) For **fig 3d**, write the expression for OCMR (OUTPUT COMMON MODE RANGE)
e) For **fig. 3g**, write the expression for ICMR (INPUT COMMON MODE RANGE)
f) For **fig. 3g**, sketch and label common mode half circuit.
g) For **fig. 3h**, write the expression for output voltage swing
h) For **fig. 3f**, write the expression for Gm
i) For **fig. 3e**, write the expression for resistance at node Vout
j) For **fig. 3d**, write the expression for low frequency small signal voltage gain.
k) Re-evaluate part (j) if voltages Vin and Vb are swapped
l) Identify type of feedback in **fig 3i**
m) For **fig. 3j**, write the expression for voltage swing in differential mode of operation
n) Write the expression for feedback factor β in **fig 3i**.
o) For **fig 3h**, write the expression for dc voltage at node Vout
p) For **fig. 3e**, write the expression for Gm.

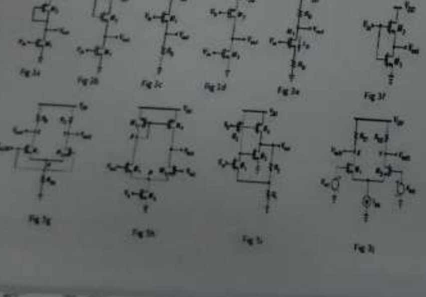


Fig 3a Fig 3b Fig 3c Fig 3d Fig 3e Fig 3f
Fig 3g Fig 3h Fig 3i Fig 3j

[30]

Q4.(a):): In the system shown in figure 4 a , the System on Chip and peripheral circuitry operates at different voltage levels. In order to make signals level compatible with each other interface circuits are required as shown in given figure.

Assume I_{bias}= 0.5mA is the current flowing through each interface circuit also values of $\mu_N C_{ox}$ =134 μA/V² $\mu_P C_{ox}$=38 μA/V², Vt_n= 0.7V, Vt_p= - 0.7V, V_{dd}=3V. Wherever required, assume basic current mirror load with V_{ov}=0.3V.

a) Choose the appropriate amplifier circuits as per the requirement shown in figure. Write its name.
b) Replace the both interface circuits blocks with the transistor level implementations of amplifier.
c) Now design your chosen amplifier as per the specification required.

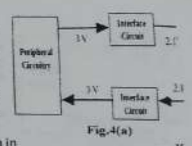
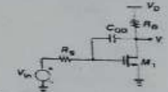


Fig.4(a)



Q4.(b): For the circuit given in figure 4(b), consider following parameters : λ=0, Rs=200Ω, Cgs=250fF, Cgd... Cdb=100fF, g_m=150 Ω⁻¹, R_D=2KΩ.

a) Calculate, low frequency voltage gain (Vout/Vin) of the amplifier
b) Also, calculate the number of poles and zeros frequency intuitively, and identify the dominant pole
c) Now, plot the frequency response (both magnitude and phase plots) up-to 10^{15} rad/s of the amplifie... appropriate labeling.(not to the scale but qualitatively correct)

[20]

Q5: The amplifier circuit shown in **figure 5**, has the current source I_{SS} of 200μA with very high R_{SS} resistance. It requires the minimum voltage of 0.2 V. Given: $(W/L)_{1-10}$ = 100, calculate the values of following using the intuitively (Assume γ=0) , I (M_9)= 100uA
(i) Input common mode range (ICMR)
(ii) Overall output voltage swing at V_{out1}.
(iii) Overall low frequency differential voltage gain (Ad=V_{out2}/V_{in}) using intuitive method.
(iv) Assuming the unity feedback, determine frequency (in rad/sec) for which phase margin is 45°.
(Given: The sum of all parasitic capacitances at node 'V_{out1}' is C_{Vout1}=200 fF, sum of all parasitic capacitances at node x/ x' or y/y' node is 10 fF, the external capacitance C_L=500 fF). Also dra... approximate Bode magnitude plot assuming a 2 dominant pole system (not to the scale but qualitatively correct
(v) If a compensating capacitor of C_c=0.5 nF , is connected between nodes 'V_{out1}' & node 'V_{out2}' then find the new locations of dominant poles of part (iv).
(vi) At the frequency obtained in part (iv), determine the new pahse angle

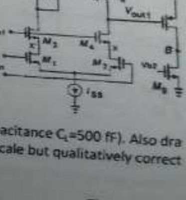


Fig.5

[20]